



Imagining Trajectories for Anomaly Detection in Reinforcement Learning from Images

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Abstract

Detecting anomalous inputs is a critical prerequisite for the safe deployment of reinforcement learning (RL) agents in real-world environments. Although anomaly detection (AD) has been extensively studied in other domains, its application to reinforcement learning remains challenging due to high-dimensional sensory observations and complex temporal dependencies. Existing approaches in this setting are limited and often rely on access to internal representations of trained agents, creating an undesirable coupling between policy and safety mechanisms. In this work, we propose ITRM, a novel approach to anomaly detection in visual reinforcement learning that is fully agent-agnostic and does not require access to policy internals. Our method is based on the observation that deviations from nominal environment dynamics can be identified through discrepancies in the latent representations of a learned world model. Specifically, we leverage predictive components of a recurrent state-space model to generate deterministic latent embeddings that serve as normative references for anomaly detection. Anomalies are detected by comparing predicted latent features against a reference set of nominal embeddings using a similarity-based criterion. Operating in the world model's latent space allows the detector to capture semantically meaningful deviations. Extensive empirical evaluations and ablation studies demonstrate that the proposed approach achieves strong detection performance. On the Anomaly-Gym benchmark, our method outperforms existing baselines, achieving an average AUROC of 0.853 and an FPR95 of 0.279.

Keywords Anomaly detection · Reinforcement learning · Out-of-distribution detection

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1 Introduction

Reinforcement learning (RL) agents deployed in real-world environments inevitably face deviations from the conditions observed during training (Amodei et al., 2016). These deviations may arise from sensor failures, actuator degradations or external perturbations, and can lead to severe performance degradation or unsafe behavior (Haider et al., 2021). Detecting such environment anomalies online – and without prior knowledge of their form – is therefore a critical prerequisite for reliable and robust RL systems. In this work, we focus on anomaly detection in high-dimensional RL environments with image-based observations. We consider a post-training monitoring setting in which the policy is fixed and anomaly detection is performed online to identify deviations in environment conditions during deployment.

Anomaly detection (AD) and out-of-distribution detection (OOD-D) have been extensively studied in supervised and unsupervised learning settings, with comprehensive surveys covering statistical, deep learning-based, and representation-driven approaches across a wide range of domains (Chandola et al., 2009; Ruff et al., 2021; Yang et al., 2024; Chalapathy and Chawla, 2019).

A common approach to anomaly detection relies on modeling the distribution of observations and identifying deviations at test time (Sakurada and Yairi, 2014; Roth et al., 2022). In RL settings, however, observations alone provide an incomplete view of the problem. They conflate transition dynamics, agent behavior, and observation noise, making it difficult to attribute deviations to genuine changes in the agent's environment. Effective anomaly detection in RL therefore requires representations that capture temporal structure and latent dynamics, rather than treating observations as independent samples.

World models (Ha and Schmidhuber, 2018) provide a natural foundation for addressing this challenge. By learning compact latent representations that summarize environment dynamics over time, world models separate observation-level variability from the underlying system behavior and impose predictive structure on sequential data. The resulting latent space offers a semantically meaningful domain in which deviations can be detected when predicted transitions no longer align with previously observed behavior.

In this work, we formulate anomaly detection in RL as the problem of identifying deviations from nominal environment conditions encoded in a learned world model. We employ a recurrent state-space model (Hafner et al., 2019) trained under a nominal MDP to learn latent representations of environment dynamics and characterize normal behavior through transition-model features in latent space. At test time, predicted latent features are compared against a reference set of nominal features using a similarity-based criterion, producing an online anomaly score without requiring anomaly-specific supervision. We refer to this approach as Imagined Trajectory Representation Matching (ITRM): representations from imagined trajectories are matched to those observed under normal conditions. Figure 1 provides an overview of the proposed approach.

Our contributions are threefold: (i) We introduce an anomaly detection mechanism operating in the latent space of a learned world model, decoupled from any policy learning. (ii) We systematically analyze key modeling choices for latent-space anomaly detection, including representation selection, distance metrics, and stochastic versus deterministic state propagation. (iii) We provide an extensive empirical evaluation assessing overall discriminative performance, time-critical detection behavior, and cross-policy transfer capabilities.

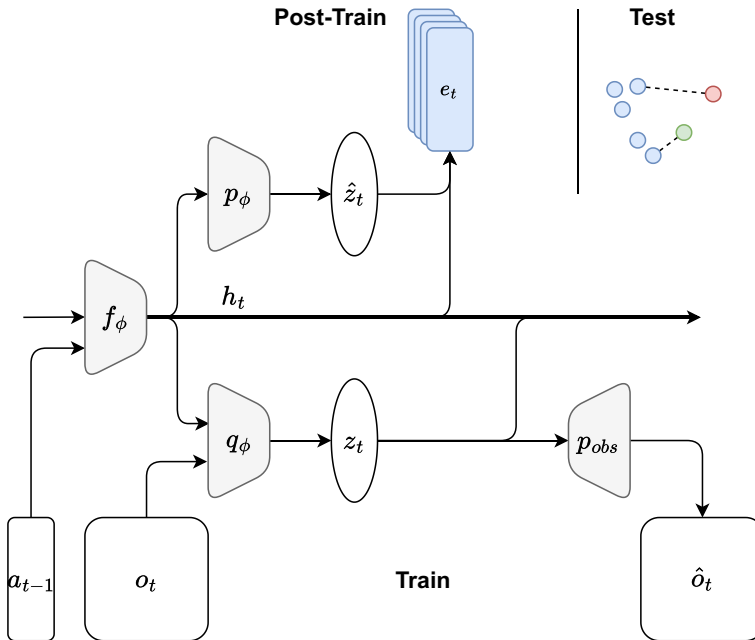


Fig. 1 Illustration of our method. The prior and posterior are optimized via the loss from Eq. 6. Post training, normal embeddings are computed and stored using the recurrent and prior models. At test-time, new embeddings are compared against stored embeddings

The code for our method and experiments is available at: <https://github.com/FraunhoferIKS/ITRM> https://anonymous.4open.science/r/itrm_ecml-21BE

2 Background and Problem Formulation

2.1 Markov Decision Process

We formulate reinforcement learning for visual environments as a *partially observable Markov decision process* (POMDP) defined by the tuple

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \mathcal{O}, \Omega),$$

where $\mathcal{S}$ denotes latent states, $\mathcal{A}$ the (continuous) action space, $P(s' | s, a)$ the transition distribution, $r(s, a)$ the reward function, $\mathcal{O}$ the space of high-dimensional observations (e.g., images), and $\Omega(o | s)$ the observation model.

At discrete time steps $t \in \{1, \dots, T\}$, the environment transitions as $s_{t+1} \sim P(\cdot | s_t, a_t)$. The agent receives an observation $o_t \sim \Omega(\cdot | s_t)$ and reward $r_t = r(s_t, a_t)$, selects actions via $a_t \sim \pi(a_t | o_{\leq t})$, and aims to maximize the expected return $\mathbb{E} \left[\sum_{t=1}^T r_t \right]$.

2.2 Anomaly Detection in Reinforcement Learning

We follow (Ruff et al., 2021) for a general definition of anomalies. Let $\mathcal{X} \subseteq \mathbb{R}^D$ denote the data space corresponding to a particular task, and let P be a probability distribution defined over $\mathcal{X}$. We define *normality* by a specific distribution P^+ over $\mathcal{X}$. An observation $x \in \mathcal{X}$ is considered an *anomaly* if it lies in a low-probability region under P^+ :

$$A = \{x \in \mathcal{X} \mid p^+(x) \leq \epsilon\}, \quad (1)$$

where p^+ is the probability density function (pdf) of P and $\epsilon \geq 0$ is some threshold. Consequently, P^- denotes the anomaly distribution and we assume that it exists on $\mathcal{X} \subseteq \mathbb{R}^D$.

In RL, the fundamental data-generating process is the interaction between policy and MDP. Throughout this work, the policy is assumed fixed after training; detected anomalies therefore correspond to changes in the environment rather than policy-induced behavioral shifts.

Hence, anomalies can be described via perturbations to individual components of the MDP, e.g., perturbations to the state space, the action space, or the transition dynamics (Haider et al., 2021). Let $\Gamma : \mathcal{M} \mapsto \mathcal{M}$ represent a perturbation applied to one or more components of an MDP. The goal of AD in RL is to determine whether a given sample x originates from the nominal MDP, denoted $\mathcal{M}^+ \equiv \mathcal{M}$, or from a perturbed (anomalous) MDP, $\mathcal{M}^- \equiv \Gamma(\mathcal{M})$. In this context, anomalies in RL are defined as:

$$A = \{x \in \mathcal{X} \mid p_{\mathcal{M}^+}(x) \leq \epsilon\}, \quad (2)$$

where x is either a single observation o_t , a single transition (o_t, a_t, o_{t+1}) or a sequence of transitions $\tau = (o_t, a_t, o_{t+1}, a_{t+1}, \dots, a_{n-1}, o_n)$. We assume a semi-supervised setting where only normal episodes are available during training and test data consist of both normal and anomalous transitions.

While we occasionally refer to related work under the umbrella of out-of-distribution (OOD) detection, we use the term anomaly detection to emphasize that our focus is on detecting general deviations in the underlying MDP, rather than purely perceptual input shifts.

3 Related Work

Anomaly and out-of-distribution (OOD) detection has been studied extensively across machine learning domains, but comparatively few methods address the reinforcement learning (RL) setting, where anomalies correspond to changes in the MDP, rather than only input shifts. This distinction is critical: in RL, anomalies often arise from modifications of the underlying Markov decision process (MDP), such as altered transition dynamics or environment parameters, and may not manifest as obvious perceptual outliers.

In the following, we review the most relevant work, with emphasis on online detection and methods applicable to high-dimensional, continuous-control environments.

Anomaly and OOD Detection in Reinforcement Learning Early work on anomaly and OOD detection in RL primarily relied on uncertainty estimates derived from the agent's policy or value function. UBOOD (Sedlmeier et al., 2019), for example, utilizes epistemic uncertainty of a policy network as an anomaly signal, showing that unfamiliar states induce elevated uncertainty. Zhang et al. (2024) introduce a distance-based anomaly detection framework that operates on class-conditional feature distributions extracted from the policy network, where anomalies are identified by measuring the distance of a state's representation to the corresponding action-specific class centroid. However, such approaches require access to agent internal representation and detection is tightly coupled to a specific policy. This makes them generally unsuitable for different RL algorithms and detection performance is coupled to the performance of the rollout policy.

A second line of work focuses on detecting anomalies via explicit modeling of environment dynamics. Methods such as RIQN (Danesh and Fern, 2021) and PEDM (Haider et al., 2023) detect anomalies by measuring prediction error or ensemble disagreement under learned transition models. These approaches are designed to capture MDP shifts directly, but they assume access to low-dimensional, fully observable state vectors. As a result, they do not extend to partially observable, image-based environments where dynamics must be inferred from high-dimensional observations.

World model-based anomaly detection has recently emerged as a promising alternative for bridging this gap. Dreamer-KLB and Dreamer-PPM (Zollicoffer et al., 2025) rely on internal signals of a Dreamer-style recurrent state-space model (RSSM) to detect novelty using latent KL-divergence. While these methods demonstrate that world models can be effective for anomaly detection, they rely on online rollouts with the Dreamer agent, and are coupled to agent-internal representations.

LDM (Haider et al., 2026) performs anomaly detection in image-based environments using latent dynamics modeling with an observation decoder. Anomalies are identified via image reconstruction error under the learned dynamics model. This enables application to high-dimensional, partially observable inputs, but the detection signal remains tied to observation-space reconstruction quality, making it sensitive to perceptual noise, decoder capacity, and visually irrelevant variations.

Deep Anomaly Detection Deep anomaly detection adapts classical one-class, reconstruction, and density-estimation ideas to learned nonlinear feature spaces. Deep SVDD (Ruff et al., 2018) trains an encoder to map normal samples into a compact hypersphere, scoring anomalies by their distance to the center. DAGMM (Zong et al., 2018) jointly optimizes an autoencoder and a Gaussian mixture model, enabling density-based scoring over reconstruction and latent features. Normalizing flows (Dinh et al., 2017) provide exact likelihood estimation via invertible mappings, yielding a principled anomaly score. These methods form the conceptual basis for the domain-specific approaches discussed below.

Visual Anomaly Detection Visual anomaly detection has been extensively studied in classification settings, where the goal is often framed as OOD detection—rejecting samples that lie outside the class label space. Autoencoder-based approaches detect anomalies via reconstruction error (Sakurada and Yairi, 2014). Distance-based variants perform kNN detection in learned latent spaces or in feature spaces of pre-trained classification networks (Bergman et al., 2020; Sun et al., 2022). More recent methods leverage self-supervised representa-

tions, such as patch-level kNN matching using DINO features (Roth et al., 2022) or nearest-neighbor detection in CLIP vision–language embedding space (Radford et al., 2021).

Video anomaly detection extends static visual methods by incorporating temporal consistency, typically detecting anomalies via elevated frame reconstruction or prediction error under learned normal dynamics (Lotter et al., 2016). While effective for short-horizon motion irregularities, these methods remain observation-centric and do not explicitly model latent environment dynamics.

Time Series Anomaly Detection Time-series anomaly detection (TSAD) addresses the detection of deviations in sequential data and has been studied extensively using both classical statistical and deep learning–based methods (Lai et al., 2021). However, most TSAD methods are designed for low- to moderate-dimensional multivariate time series, with typical inputs consisting of vectors of sensor or system measurements (e.g., OmniAnomaly (Su et al., 2019), TranAD (Tuli et al., 2022), AnomalyTransformer (Xu et al., 2022)). These methods do not scale directly to high-dimensional observations such as images and become intractable.

Summary In summary, visual anomaly detection methods primarily focus on static perceptual novelty and largely ignore temporal structure, making them unsuitable for detecting anomalies that manifest through changes in environment dynamics. Time-series anomaly detection methods explicitly model temporal dependencies but are designed for low- to moderate-dimensional signals and do not scale to high-dimensional image observations. This motivates our approach that aligns world model representations with distance-based anomaly detection techniques that have proven effective in modern visual OOD detection while remaining applicable to high-dimensional, partially observable reinforcement learning environments. Our approach decouples policy learning from anomaly detection by using a world model solely to learn compact latent representations of nominal environment dynamics. Rather than relying on agent-internal signals or observation-space reconstruction error, anomalies can be identified via implicit local density estimation over latent embeddings, for example by measuring similarity to a reference set of nominal features.

4 Method

Following the definition of anomalies in Section 2.2, the goal of AD in RL is to identify low-probability regions under the support of the data-generating distribution $\mathcal{M}^+$, while only samples from the nominal MDP $\mathcal{M}^+$ are available for training. In high-dimensional visual environments with action-conditioned dynamics, modeling this distribution solely in observation space ignores the sequential and control-dependent structure of the problem. We therefore approach anomaly detection indirectly by learning a latent generative model of the environment dynamics and using the induced latent state distribution as a proxy for the support of $P_{\mathcal{M}^+}$. Anomalies are then detected as deviations from this learned latent distribution.

This formulation naturally decomposes the problem into three components: (i) learning a latent representation that captures the nominal environment dynamics, (ii) selecting latent variables whose distribution reflects normality under $\mathcal{M}^+$ and (iii) defining a decision rule that identifies deviations from this distribution at test time. In this work, we consider a

recurrent state-space model as the latent generative model (i), derive representations from its hidden state and prior (ii), and define anomaly scores by comparing these representations to a feature bank of nominal trajectories (iii). Figure 1 provides a summary of our method. In the following, we describe each component in detail.

4.1 Generative World Models

To approximate the support of the data-generating distribution of the normal environment in high-dimensional visual domains, we train a generative world model that captures environment dynamics in a latent state space. A generative world model is a latent variable model that defines a joint distribution over observation sequences, actions, and latent states,

$$p_{\phi}(o_{1:T}, z_{1:T} \mid a_{1:T}) \quad (3)$$

where the latent variables z_t represent abstract environment states. The model is trained exclusively on trajectories generated under the nominal MDP $\mathcal{M}^+$ and thereby induces a latent state distribution whose support approximates the support of $P_{\mathcal{M}^+}$.

In contrast to direct density estimation in observation space, this induced distribution is compact, semantically structured, and constrained by the learned temporal dynamics. As a result, trajectories generated under the nominal environment concentrate on a restricted region of latent space, while deviations from the nominal dynamics occupy lower-density regions. Anomaly detection can therefore be formulated as identifying deviations from the support of the induced latent state distribution.

In this work, we adopt a recurrent state-space (RSSM) formulation (Hafner et al., 2019), which provides a structured factorization of latent dynamics and explicitly distinguishes predictive and observation-conditioned inference.

4.2 Latent Dynamics Via Recurrent State-Space Models

Recurrent state-space models (RSSMs) define a latent-variable formulation for sequential decision-making that combines deterministic recurrent states with stochastic latent variables to model environment dynamics under partial observability. At each time step t , the RSSM maintains a deterministic recurrent state h_t , which summarizes past latent states and actions, and a stochastic latent state z_t , which captures uncertainty about the current environment state. The RSSM specifies two conditional distributions over the latent state: a predictive prior which encodes the model's belief about the latent state before observing o_t , and an observation-conditioned posterior, which incorporates information from the current observation. This explicit separation between predictive and observation-conditioned inference is central to our anomaly detection formulation and will be leveraged in the following sections. In summary, the RSSM defines the following components:

$$\begin{array}{ll} \text{Recurrent model:} & h_t = f_{\phi}(h_{t-1}, z_{t-1}, a_{t-1}) \\ \text{Prior (transition model):} & \hat{z}_t \sim p_{\phi}(\hat{z}_t \mid h_t) \\ \text{Posterior (representation model):} & z_t \sim q_{\phi}(z_t \mid h_t, o_t) \end{array} \quad (4)$$

As an additional learning signal,¹ observations are reconstructed via

$$\text{Observation model: } o_t \sim p_{\text{obs}}(o_t | h_t, z_t) \quad (5)$$

All components are trained jointly by maximizing the evidence lower bound:

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi(z|h,o)} \left[\underbrace{\sum_{t=1}^T -\log p_{\text{obs}}(o_t | h_t, z_t)}_{\text{Image Loss}} + \beta \underbrace{\text{KL}(q_\phi(z_t | h_t, o_t) || p_\phi(\hat{z}_t | h_t))}_{\text{KL Loss}} \right]. \quad (6)$$

During training, the prior is aligned with the observation-conditioned posterior through the KL term in eq. 6. At deployment, the prior is used only to compute the latent feature used for anomaly detection. In contrast, the recurrent state update remains observation-conditioned: after receiving o_t , we infer the posterior z_t and this posterior latent is used to update the next hidden state h_{t+1} . Therefore, the hidden state is re-anchored to the current observation at every step, which prevents cumulative drift. In the following, we discuss which of the RSSM's latent components are most suitable for AD.

4.3 Latent Representations for Detecting Distribution Shift

The RSSM factorizes environment dynamics into deterministic and stochastic components through distinct latent variables. Each component captures a different source of variability and thus induces a different notion of normality. For anomaly detection under the nominal MDP $\mathcal{M}^+$, we analyze three representations corresponding to these components.

Deterministic recurrent state The deterministic recurrent state $h_t = f_\phi(h_{t-1}, z_{t-1}, a_{t-1})$ models the predictable, deterministic structure of the environment by aggregating past observations and actions. It provides a stable, temporal representation that captures deviations in the latent dynamics, even though it does not correspond to an explicit probability distribution.

Posterior latent state The posterior latent state $z_t \sim q_\phi(z_t | h_t, o_t)$ captures stochastic components of the environment while conditioning on the current observation. This representation integrates perceptual information with latent uncertainty, making it sensitive to observation-level irregularities as well as stochastic variations not captured by the deterministic dynamics.

Predictive prior The predictive prior $\hat{z}_t \sim p_\phi(\hat{z}_t | h_t)$ captures stochastic components of the dynamics through the model's prediction of future latent states, without conditioning on the current observation. As a result, deviations in this representation reflect mismatches in learned transition dynamics, with less dependence on the current observation than posterior-based features.

¹ Some works also predict the reward r_t and the discount factor γ_t which are useful for planning in latent space. Since we don't require planning we refrain from this here but note that both could also be included as additional learning signals.

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Combined deterministic–stochastic representation In our final method, we combine the deterministic recurrent state with the predictive prior. This joint representation integrates the stable temporal context provided by h_t with the dynamics-level stochastic component encoded in $\hat{z}_t$. The resulting embedding captures structured deviations in the predictable dynamics together with shifts in the model’s predicted latent state.

Sampling Features & Rollouts The predictive prior and posterior are parameterized as explicit probability distributions, from which latent features are instantiated by sampling. In addition to this approach, we also consider a deterministic alternative that uses summary statistics of the predicted distribution, such as the mean (for Gaussian latents) or the mode (for categorical latents). This choice reduces sampling-induced variance and yields more consistent representations across time steps and episodes. The same distinction applies to temporal propagation: instead of forwarding a sampled latent to the next time step, we additionally evaluate expectation-based rollouts that propagate the latent mean or mode. This formulation allows us to disentangle the effects of stochastic instantiation from those of the underlying latent representation.

4.4 Anomaly Detection in Predictive Latent Space

To perform anomaly detection at test time, we follow a simple, transparent, and training-free nonparametric local density estimation approach based on k-nearest neighbors. kNN is a natural choice here: it makes minimal distributional assumptions, is straightforward to integrate, and allows us to isolate the effect of the learned representation from that of a more complex scoring model. Deterministic deep features e_t are directly extracted from the RSSM. Let

$$\mathcal{B} = \{e^{(i)}\}_{i=1}^N, \quad (7)$$

define a feature bank computed from normal training trajectories, which serves as an empirical approximation of the latent state distribution induced by the nominal MDP.

Given a test-time feature e_t , we compute an anomaly score based on the average distance to its k nearest neighbors in the feature bank:

$$d_k(e) = \frac{1}{k} \sum_{e' \in \text{NN}_k(e; \mathcal{B}, d)} d(e, e'), \quad (8)$$

where $d(\cdot, \cdot)$ denotes a distance function on the feature space (e.g., Euclidean distance, cosine similarity) and $\text{NN}(e, \mathcal{B}, d)$ is the set of the k-nearest neighbors of e in $\mathcal{B}$ under this distance.

The resulting score serves as a proxy for inverse local density in the predictive latent space: features corresponding to nominal states admit close neighbors, while features arising from distributional shift are isolated and yield larger distances.

A test sample is classified as anomalous if its anomaly score exceeds a threshold

$$e \in \mathcal{A} \quad \text{if} \quad d_k(e) > \vartheta. \quad (9)$$

The threshold ϑ is selected based on the empirical distribution of anomaly scores on nominal validation data.

5 Evaluation

5.1 Experimental Setup Overview

Datasets & environments: We empirically evaluate our method on Anomaly-Gym, a recently introduced benchmark suite for anomaly detection in reinforcement learning under controlled environment shifts (Haider et al., 2026). The benchmark covers 10 different continuous-control tasks (3 navigation tasks, 3 robotic control tasks, 3 robotic manipulation tasks and 1 autonomous driving task). For each environment, there are 12 different anomaly types, affecting observations, dynamics, or actions, with random onset times and different severity levels.

We consider a semi-supervised anomaly detection setting, where all models are trained exclusively on nominal data and evaluated on both nominal and anomalous data. For each task, we use 100 normal episodes for training and 10 normal episodes for validation (with different initial seeds), following the benchmark protocol. The final performance is assessed on a dedicated test set, which contains 100 episodes for each combination of anomaly type and anomaly strength. In anomalous test episodes, the anomaly onset time is randomized: $t_\alpha \sim \mathcal{U}(0, H)$ (randomized onset). The timesteps $[t_0, \dots, t_{\alpha-1}]$ are labeled as nominal, whereas $[t_\alpha, \dots, t_H]$ are labeled as anomalous. We denote by $\tau_{\pi, \mathcal{M}}$ a trajectory generated by executing policy π in environment $\mathcal{M}$. The resulting test dataset is balanced in expectation and given by

$$\mathcal{D}_{\text{test}} = \{\tau_{\pi, \mathcal{M}^+}^{[0:\alpha-1]}\}_{n=1}^N \cup \{\tau_{\pi, \mathcal{M}^-}^{[\alpha:H]}\}_{n=1}^N. \quad (10)$$

Metrics We evaluate our method using established anomaly detection metrics. We report the Area Under the Receiver Operating Characteristic curve (AUROC) (Fawcett, 2006) as a measure of overall discriminatory performance, and the False Positive Rate at 95% True Positive Rate (FPR95) to analyze performance at low false-positive operating points. Since we analyze detection latency explicitly in Section 5.3, we restrict overall detection results to these two metrics. Detailed results with all metrics are also provided in Appendix B.

Model training & hyper-parameters The recurrent state-space model (RSSM) is trained in a self-supervised manner on nominal episodes only, following the architecture of (Hafner et al., 2019) with minor modifications to accommodate higher-resolution observations. We train for 200 epochs using full-length episodes (one episode per batch), a learning rate of 2×10^{-4} , and a KL-loss scaling factor of $\beta = 1.0$. After training, the world model is frozen and used solely for feature extraction during evaluation. Full architectural details and hyper-parameters are provided in Appendix A.1.

Baselines We compare our approach against a diverse set of baselines from both computer vision and reinforcement learning. From the computer vision domain, we include reconstruction- and feature-based methods commonly used for visual anomaly detection: image reconstruction with an autoencoder (**AE-Rec**), kNN distance using pre-trained ResNet features (**RES-KNN**) similar to (Sun et al., 2022), predictive coding via **PredNet**

(Lotter et al., 2016), patch-based kNN detection using DINO features (**DINO-Patch**) following (Roth et al., 2022), and kNN detection using CLIP embeddings (**CLIP-Sim.**) similar to Sun et al. (2022).

From the reinforcement learning domain, we compare against **LRDM** (Haider et al., 2026) as well as **RSSM-KLD** and **RSSM-PPM** (Zollicoffer et al., 2025). The latter leverage internal representations of a world model, which we train analogously to the RSSM of our method via self-supervised learning on normal data, using the same world model architecture and training protocol wherever applicable. Methods requiring discrete action spaces or access to a pre-trained PPO agent (Sedlmeier et al., 2019; Zhang et al., 2024) are not applicable to the continuous-control environments considered in the Anomaly-Gym dataset and are therefore excluded.

5.2 Overall Detection Performance

We evaluate overall anomaly detection performance using AUROC and FPR95, aggregated across all tasks, anomaly types, and two anomaly strengths (tiny and strong) in the benchmark. Figure 2a and Fig. 2b report the corresponding results.

Our method achieves the highest average AUROC, with a mean score of 0.853, indicating strong overall discriminatory performance across heterogeneous environments and anomaly conditions. Among the baselines, RSSM-KLD performs best, while other approaches exhibit substantially lower average AUROC. Across all methods, we observe considerable variance in performance, reflecting the diversity of tasks and the wide range of anomaly severities present in the benchmark.

Consistent with the AUROC results, our method also attains the lowest average FPR95, with a mean value of 0.279 (Fig. 2b). However, FPR95 exhibits high variance across anomaly conditions for all evaluated methods, including ours. This highlights the difficulty of selecting a single operating threshold that yields reliable low false-positive rates across diverse environments, a challenge we further investigate in the detection time analysis in Section 5.3.

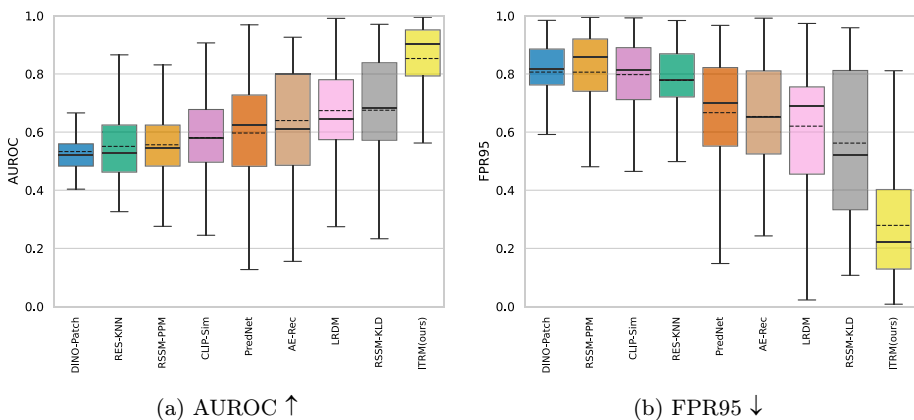


Fig. 2 AUROC and FPR95 across all environments, anomalies, and strength levels. Solid line: 50th percentile, dashed line: mean

Notably, the three best-performing methods in this evaluation are all based on learned world models, underscoring the effectiveness of model-based representations for anomaly detection. Among these, RSSM-KLD employs the same underlying network architecture as our approach, yet achieves substantially lower aggregate performance. Our method consistently outperforms the remaining world model-based baselines, indicating that the performance gains cannot be attributed to architectural capacity alone but rather to differences in representation choice and scoring. We analyze the factors contributing to these differences in detail in Section 5.5.

To complement aggregated results, Fig. 3 reports per-environment AUROC for ITRM and the two strongest baselines (LRDM and RSSM-KLD). ITRM is particularly strong and stable on the SAP-Goal tasks, CAR-LaneKeep, MJC-HalfCheetah, and MJC-Reacher3D, where it achieves high median AUROC with relatively small spread. The most challenging environments remain URM-PnP, URR-Reach and MJC-CartpoleSwingup, all of which show lower performance and higher variability across methods overall. In the latter, LRDM even achieves a higher mean AUROC than ITRM, although with a smaller median and higher variance. Overall, the per-environment results indicate that the gains are broad rather than driven by only a small subset of tasks.

5.3 Anomaly Thresholding and Detection Time

While threshold-free metrics such as AUROC characterize overall discriminatory performance, they do not capture how anomaly scores behave over time once a decision threshold is applied. In practical settings, detection time is critical for understanding the operational utility of anomaly detection methods beyond aggregate ranking metrics.

In this section, we study how quickly anomalies are detected after onset under different thresholding strategies. For clarity, we report detection time results for our method and the strongest competing baselines (LRDM and RSSM-KLD).

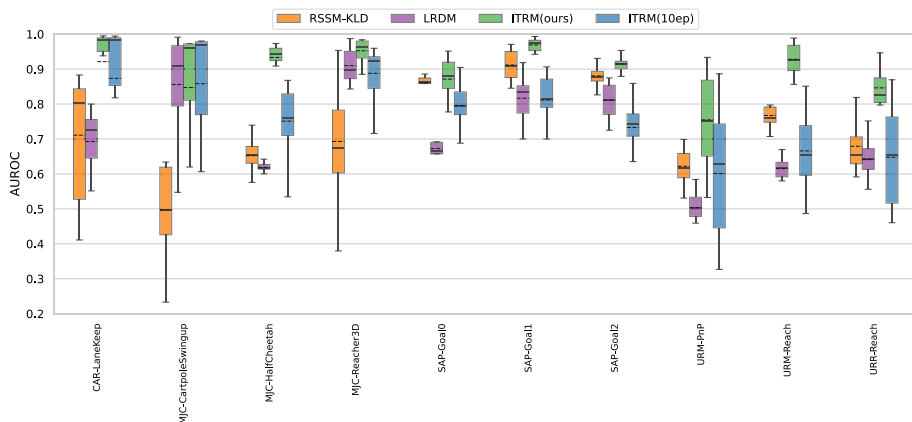


Fig. 3 Per-task results (AUROC) of our method and the two best baselines on strong anomalies. ITRM(10ep) is trained with only 10 episodes instead of 100

5.3.1 Thresholding Protocols

We evaluate the following thresholding strategies, all calibrated exclusively using nominal validation data:

- **3 σ threshold (3 σ).** A fixed global threshold defined as

$$\vartheta = \mu + 3\sigma,$$

where μ and σ denote the mean and standard deviation of anomaly scores on nominal validation trajectories.

- **Quantile threshold (95-ID)** A fixed global threshold chosen as the 95th percentile of nominal validation scores:

$$\vartheta = \text{Quantile}_{0.95}(\{\nu_t \mid t \in \mathcal{D}_{\text{val}}^+\}).$$

where ν_t is the anomaly score.

- **Adaptive EMA threshold (EMA)** An online threshold updated as

$$\mu_t = (1 - \lambda)\mu_{t-1} + \lambda\nu_t, \quad \sigma_t^2 = (1 - \lambda)\sigma_{t-1}^2 + \lambda(\nu_t - \mu_t)^2,$$

$$\vartheta_t = \mu_t + k\sigma_t,$$

where μ_t and σ_t are exponentially smoothed statistics of past anomaly scores.

5.3.2 Detection Time Metrics

Let t_α denote the anomaly onset time in an episode. Detection delay is defined as

$$\Delta t = \min\{t \geq t_\alpha \mid \nu_t > \vartheta\} - t_\alpha,$$

where ϑ is the chosen threshold. Episodes in which no post-onset threshold crossing occurs are counted as missed detections.

We report the following metrics aggregated across all tasks, anomaly types, and anomaly strengths:

- **Median Detection Delay (Med. Delay)**, computed over successfully detected episodes;
- **Detection after K steps (D_K)**, the fraction of anomalous episodes detected within $K \in \{5, 10, 20\}$ timesteps after onset;
- **Miss Rate (MR)**, the fraction of anomalous episodes with no post-onset detection; and
- **Early Detection Rate (EDR)**, defined as the fraction of episodes in which the threshold is crossed before anomaly onset.

5.3.3 Detection Time Results

Table 1 summarizes detection time performance under the different thresholding strategies. Across all thresholds, our method achieves the lowest median detection delay and the highest D_K scores, indicating faster detection after anomaly onset. In particular, improvements are most pronounced for small K , highlighting more timely detection in the early stages following a shift.

Despite these gains, detection time performance exhibits substantial variance across anomaly conditions for all methods. Miss rates remain non-negligible, especially under fixed global thresholds, underscoring the difficulty of selecting a single operating point that performs reliably across heterogeneous environments and anomaly severities. Early detections are observed for all methods, reflecting an inherent trade-off between rapid detection and false alarms.

Adaptive EMA thresholding generally reduces miss rates compared to fixed thresholds but increases early detection rates, illustrating the sensitivity of online detection performance to threshold dynamics. Overall, these results highlight that strong aggregate discrimination does not necessarily translate into fast or reliable detection under a fixed threshold, motivating careful consideration of thresholding strategies in practical deployments.

For a more nuanced view on anomaly thresholds, we plot anomaly score over time in Fig. 4, together with different thresholding strategies. Specifically, we consider two episodes from *MJC-Reacher3D*. Figure 4a shows a positive example, where the anomaly score remains below thresholds until shortly after anomaly onset, when it explodes. Figure 4b, on the other hand visualizes that in some cases, the anomaly score can surpass thresholds even before anomaly occurrence.

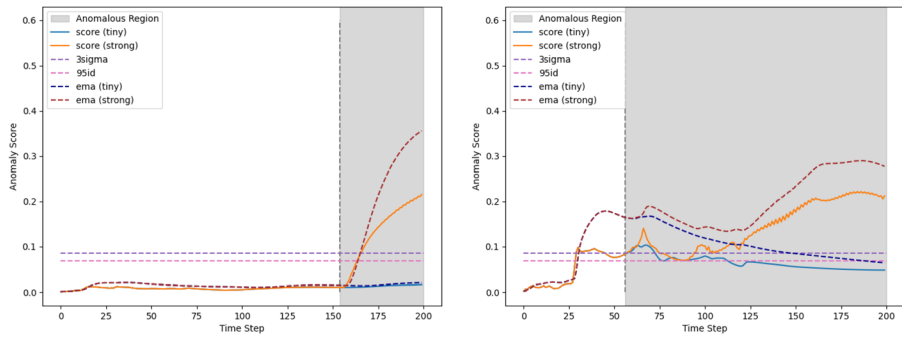
5.4 Cross-Policy Transfer

One advantage of world model features is general, policy-independent representations. Provided the training policy induces sufficient coverage of nominal environment dynamics,

Table 1 Detection performance reported as mean $\pm$ standard deviation under different thresholding strategies

Detector	Med Del ($\downarrow$)	D_5 ($\uparrow$)	D_{10} ($\uparrow$)	D_{20} ($\uparrow$)	MR ($\downarrow$)	EDR ($\downarrow$)
<i>3σ Threshold</i>						
LRDM	53.11 $\pm$ 51.14	0.10 $\pm$ 0.20	0.19 $\pm$ 0.29	0.26 $\pm$ 0.31	0.67 $\pm$ 0.35	0.22 $\pm$ 0.31
RSSM-KLD	32.41 $\pm$ 32.43	0.14 $\pm$ 0.24	0.26 $\pm$ 0.31	0.37 $\pm$ 0.33	0.56 $\pm$ 0.39	0.31 $\pm$ 0.37
ITRM	29.16 $\pm$ 31.48	0.24 $\pm$ 0.33	0.33 $\pm$ 0.39	0.42 $\pm$ 0.40	0.57 $\pm$ 0.42	0.21 $\pm$ 0.31
<i>95-ID Threshold</i>						
LRDM	39.16 $\pm$ 38.64	0.16 $\pm$ 0.17	0.21 $\pm$ 0.19	0.30 $\pm$ 0.21	0.38 $\pm$ 0.31	0.40 $\pm$ 0.30
RSSM-KLD	22.52 $\pm$ 20.53	0.23 $\pm$ 0.30	0.35 $\pm$ 0.35	0.47 $\pm$ 0.34	0.40 $\pm$ 0.36	0.42 $\pm$ 0.38
ITRM	18.18 $\pm$ 21.46	0.34 $\pm$ 0.35	0.46 $\pm$ 0.37	0.55 $\pm$ 0.34	0.31 $\pm$ 0.34	0.41 $\pm$ 0.35
<i>EMA Threshold</i>						
LRDM	16.72 $\pm$ 21.89	0.38 $\pm$ 0.31	0.60 $\pm$ 0.30	0.69 $\pm$ 0.27	0.19 $\pm$ 0.23	0.50 $\pm$ 0.35
RSSM-KLD	10.97 $\pm$ 14.34	0.39 $\pm$ 0.26	0.63 $\pm$ 0.32	0.71 $\pm$ 0.27	0.15 $\pm$ 0.21	0.57 $\pm$ 0.37
ITRM	5.02 $\pm$ 3.80	0.54 $\pm$ 0.25	0.79 $\pm$ 0.22	0.84 $\pm$ 0.17	0.05 $\pm$ 0.08	0.65 $\pm$ 0.35

($\downarrow$) lower is better, ($\uparrow$) higher is better



(a) Anomaly Scores stay below threshold until shortly after anomaly onset.

(b) Anomaly Scores surpass threshold before anomaly onset.

Fig. 4 Visualization of Anomaly-Score and Thresholds over Time for two different episodes in *MJC-Reacher3D* with both tiny (blue) and strong(orange) severity

Table 2 Cross-policy transfer performance

Detector	Test Policy	AUROC	FPR95
LRDM	TQC	0.71	0.63
	SAC	0.62	0.73
	TD3	0.61	0.74
RSSM-KLD	TQC	0.72	0.52
	SAC	0.71	0.51
	TD3	0.71	0.52
ITRM	TQC	0.86	0.28
	SAC	0.73	0.44
	TD3	0.71	0.48

this allows AD in a cross-policy transfer setting. That is, training the detector using data generated with policy A, and evaluating it on data generated with policy B. We conduct this experiment on a subset of three environments (SAP-Goal0, MJC-CartpoleSwingup, URM-Reach) under strong anomaly conditions and again compare against the two strongest baselines. The detectors are trained using trajectories collected with a TQC policy and evaluated on test trajectories generated by different policies. Results are reported in Table 2.

Even in this challenging setting, our method achieves an AUROC of 0.73/0.71, although the FPR95 nearly doubles. A likely reason is that the model is trained only on the local region of the state space visited by Policy A. When deployed with Policy B, however, the agent may encounter substantially different states even under nominal conditions, which limits generalization. Compared with LRDM, ITRM remains clearly stronger in both the original and transfer settings. Interestingly, although RSSM-KLD performs substantially worse in the original setting, it degrades much less under cross-policy transfer. Even so, ITRM still achieves the lowest overall FPR95 across policies.

Overall, these results highlight that our approach is fully independent of the deployed RL algorithm and supports meaningful transfer across policies to a meaningful extent.

5.5 Ablations

To analyze the contribution of individual modeling choices, we conduct a series of ablation studies that systematically vary key components of our method. Aggregate performance metrics are reported in Table 3, while Fig. 5 provides a more fine-grained view of the resulting anomaly score distributions under nominal and anomalous conditions for *strong observation noise* anomalies in *MJC-HalfCheetah*.

Distance Metric We evaluate several distance metrics for nearest-neighbor retrieval in latent space, including ℓ_1 , ℓ_2 , cosine, and Mahalanobis distance. Overall differences in AUROC are modest across choices, indicating that performance is not overly sensitive to the specific metric. The ℓ_1 distance achieves the highest AUROC (0.853). Interestingly, although Mahalanobis distance attains comparable AUROC, it exhibits a higher false positive rate, suggesting that covariance estimation from limited nominal data can lead to overly confident deviations.

Influence of the number of neighbors We further study the impact of the number of nearest neighbors k used to compute anomaly scores. In our setting, smaller values of k perform better than larger ones, with $k=1$ giving the best overall results. This suggests that the most informative anomaly signal is highly local in the learned latent space. We attribute the degradation for larger k primarily to the limited size of the feature bank: as k increases, the score is increasingly influenced by less representative neighbors, which smooths away informa-

Table 3 Ablation Studies across scoring function, distance metric, number of neighbors, and latent components

Category	Setting	AUROC		FPR95	
		Mean	Std	Mean	Std
Scoring Function	KL	0.657	0.164	0.583	0.210
	Obs. Rec	0.719	0.147	0.546	0.223
	LOF (L1)	0.845	0.141	0.371	0.227
	LOF (L2)	0.823	0.149	0.415	0.234
	INFLO (L1)	0.852	0.138	0.329	0.221
	INFLO (L2)	0.833	0.146	0.362	0.232
Distance Metric	L_2	0.830	0.154	0.318	0.199
	Cosine	0.838	0.135	0.311	0.180
	Mahalanobis	0.842	0.140	0.358	0.224
	L_1	0.853	0.144	0.279	0.192
k Nearest Neighbors	$k = 50$	0.760	0.230	0.417	0.270
	$k = 10$	0.813	0.174	0.344	0.221
	$k = 5$	0.832	0.158	0.314	0.207
	$k = 4$	0.836	0.155	0.307	0.204
	$k = 3$	0.842	0.151	0.299	0.200
	$k = 2$	0.848	0.148	0.288	0.197
	$k = 1$	0.853	0.144	0.279	0.192
Latent Component	Prior (stoch)	0.518	0.068	0.885	0.092
	Posterior (stoch)	0.519	0.068	0.885	0.092
	Posterior (det)	0.767	0.170	0.429	0.235
	Prior (det)	0.811	0.150	0.363	0.198
	Hidden (stoch)	0.836	0.148	0.313	0.207
	Hidden (det)	0.853	0.144	0.281	0.190
	Prior $\oplus$ Hidden (det)	0.853	0.144	0.279	0.191

For all results, only the specific parameter/setting was changed relative to the final method

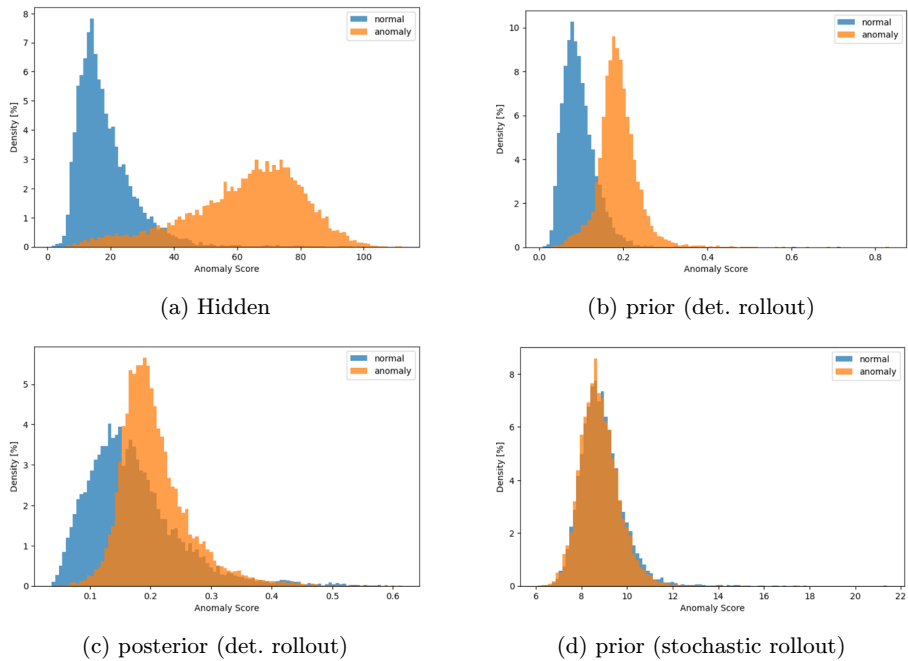


Fig. 5 Ablation studies: Distributions of anomaly scores for normal and anomalous data, with different latent components

tive local deviations. While relying on a single neighbor can in principle be more sensitive to noise or outliers, this effect appears limited in our setting due to the use of deterministic rollouts and deterministic latent features, which reduce score variance.

Latent component We also ablate the choice of latent representation used for detection. Deterministic recurrent (hidden) features perform strongly, while explicitly modeling stochastic latent components does not lead to improved detection performance (see also Fig. 5a). In the evaluated benchmark, the hidden state already captures most of the useful anomaly signal, so adding the prior does not yield a consistent gain. We attribute this to the largely predictable nature of the Anomaly-Gym environments, where stochastic effects play only a minor role and the hidden state is already highly informative. The prior remains conceptually well motivated because it captures predictive dynamics before posterior correction, but its practical benefit in this benchmark appears limited and complementary rather than consistently additive.

Comparing stochastic latents, we find that prior-based embeddings consistently outperform posterior embeddings. Since the posterior is conditioned on observations and optimized for reconstruction, it may retain more observation-specific visual detail, which could make the anomaly signal less stable. This is also evident from Fig. 5b and 5c.

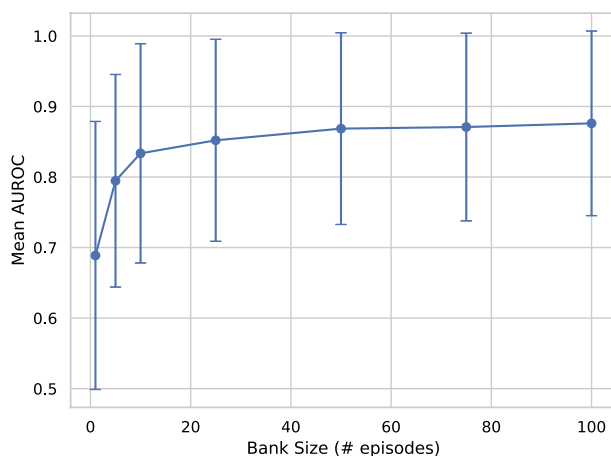
Moreover, using samples from the latent distribution performs significantly worse than using its expectation (mean or mode), due to the additional variance introduced by sampling (see Fig. 5d). A similar trend is observed for stochastic rollouts, which are slightly inferior to rollouts computed using latent expectations, even when detection is based solely on the deterministic hidden state (see Table 3: *Hidden (stoch)* vs. *Hidden (det)*).

Scoring Function We compare several alternative scoring functions using the same underlying RSSM. We evaluate the latent KL divergence between prior and posterior $KL(p_\phi \| q_\phi)$ as well as the observation-space reconstruction loss $\|o_t - \hat{o}_t\|_2$. In our experiments (see Table 3), both score below the distance-based methods in latent space. To test whether more elaborate local outlier scores provide an advantage in the same latent space, we also evaluate alternative neighbor-based scores in the same latent space, namely LOF (Breunig et al., 2000) and INFLO (Jin et al., 2006). While these methods are competitive, neither improves over the NN-based detector. This supports kNN as a strong and simple default choice in our setting.

Feature Bank Size To analyze the effect of feature-bank coverage, we additionally study detection performance across different feature-bank sizes. The results (see Fig. 6) show that AUROC improves as the bank grows, indicating that increased coverage substantially helps in the low-data regime. However, the gains become progressively smaller after approx. 10 reference episodes, suggesting diminishing returns once the bank provides sufficient coverage of nominal behavior. Overall, this analysis supports two points: the method does benefit from better coverage, but it is also reasonably robust to moderate bank sizes, since performance is already close to saturation with a relatively small bank.

World Model Capacity To analyze the effect of world model capacity on detection performance, we also conducted an ablation varying the number of learnable parameters. While this does not cover all possible forms of misspecification, it provides a controlled way to study how imperfect world models affect anomaly detection. The results (see Fig. 7a) show that very small world models exhibit lower detection performance. As capacity increases, performance improves and then largely saturates in the mid-size range, with only small differences among sufficiently large models. We further observe a visible, but not one-to-one, correspondence between detection performance and final world model test loss (see Fig. 7b), indicating that accurate nominal modeling is important, but does not fully determine anomaly separability in latent space. Overall, detection performance appears fairly robust once the world model is sufficiently large.

Fig. 6 Influence of feature bank size (all environments, strong anomalies)



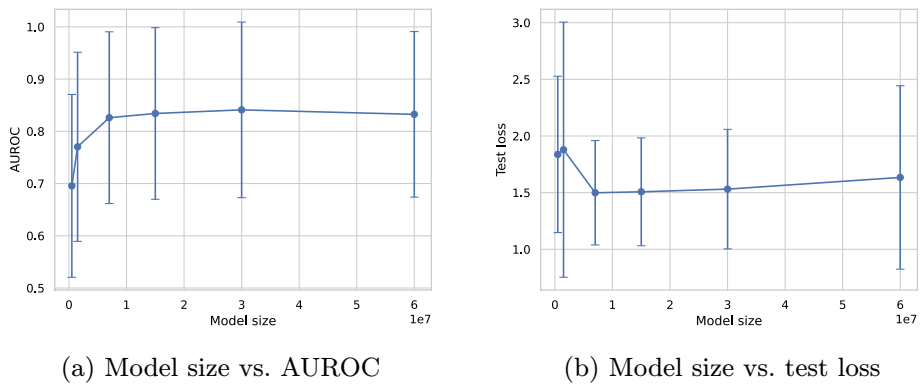


Fig. 7 Influence of model size (all environments, strong anomalies); our method uses approx. 3×10^7 model parameters

Table 4 Sub-sampling Strategies: AUROC $\pm$ std. with different fractions of data used for the feature bank

Method	5%	10%	25%	50%
episode sampling	0.725 $\pm$ 0.17	0.773 $\pm$ 0.18	0.819 $\pm$ 0.16	0.835 $\pm$ 0.15
uniform sampling	0.751 $\pm$ 0.20	0.780 $\pm$ 0.17	0.819 $\pm$ 0.16	0.839 $\pm$ 0.14
k-center subset	0.803 $\pm$ 0.15	0.825 $\pm$ 0.14	0.847 $\pm$ 0.13	0.846 $\pm$ 0.14

Influence of Nominal Data Coverage To further investigate the effect of data coverage, we additionally study a low-data regime in which the world model is trained on only 10 nominal episodes instead of 100, and evaluate the resulting per-task AUROC. The results are presented in Fig. 3 and show noticeably lower performance on almost all tasks, indicating that sufficient coverage of nominal behavior during world model training is important for reliable anomaly detection. Interestingly, for MJC-CartpoleSwingup there is no performance degradation. This reflects low trajectory diversity in that environment, such that even 10 episodes already provide sufficient coverage of the nominal dynamics.

Sub-sampling strategies To reduce the memory and computational cost of kNN scoring, we study whether the feature bank can be compressed while preserving detection performance. We evaluate three sub-sampling strategies and different fractions of the training bank. **Episode sampling** keeps a fraction of complete episodes, **uniform sampling** selects individual feature vectors uniformly at random from the flattened bank. **k-center coreset sampling** greedily builds a subset by repeatedly adding the point that is farthest from its nearest already-selected point. This strategy is used in memory-bank anomaly detectors such as PatchCore (Roth et al., 2022). Results are presented in Table 4. Overall, k-center consistently performs best across all compression levels, with the largest gains at low fractions. Notably, at only 5% of the bank, k-center achieves an AUROC of 0.803, corresponding to a degradation of only 0.05 AUROC relative to the full-bank setting. Performance largely saturates beyond 25%, suggesting that a compact, coverage-based coreset can preserve most of the useful nearest-neighbor structure.

6 Conclusions, Limitations, and Outlook

In this work, we introduced a novel approach to anomaly detection (AD) in high-dimensional reinforcement learning environments based on learned world models. By training the dynamics model post hoc, the proposed method is decoupled from policy-internal representations and can be deployed as an external monitoring module. Our central contribution is the demonstration that a deep nearest-neighbor detector operating on deterministic embeddings derived from world model latents constitutes a highly effective and robust anomaly detection strategy.

We motivated the combination of deterministic recurrent state features with deterministic statistics of the latent prior from a modeling perspective. The recurrent hidden state provides a stable summary of the environment's deterministic structure and is less susceptible to stochastic noise, while the latent prior reflects the model's predictive dynamics but is not influenced by posterior correction effects.

Although our empirical results do not show a consistent improvement from explicitly combining prior and hidden-state representations in the evaluated benchmarks, this can be attributed to the largely predictable nature of the considered environments, where deterministic dynamics dominate and are already well captured by the recurrent state alone. We nevertheless argue that this design choice is principled and likely to become important in environments with stronger stochasticity. Testing this hypothesis requires further investigation.

Extensive empirical evaluations and ablation studies demonstrate that our method, ITRM, consistently outperforms reconstruction-based baselines and alternative world model-based approaches relying on posterior representations. In particular, our results highlight the importance of deterministic components for reducing variance in anomaly scores and show that posterior-based features tend to be less reliable than prior-based ones, likely because they are more tightly coupled to observation-specific variation.

Despite these encouraging results, several limitations remain. Most notably, our experiments indicate that reliable threshold selection remains a major challenge for practical deployment. Future work could explore more robust thresholding strategies, for example by incorporating limited exposure to anomalous data or adaptive calibration schemes. Another important direction is to extend the setting to anomalies during training, both for contaminated but unlabeled data and for settings with explicit anomaly labels. Additional directions include investigating alternative world model architectures and representation learning approaches. In particular, combining pretrained visual encoders, such as DINO or CLIP, with RSSM-style dynamics models may yield more expressive and semantically meaningful latent spaces. Furthermore, a dedicated ablation isolating robustness to visual noise is outside the scope of the current evaluation and remains an important direction for future work.

Finally, exact nearest-neighbor search introduces a potential scalability bottleneck, since both memory and query cost grow linearly with the size of the feature bank. In our current

setting, however, this cost remains manageable: the bank is built from 100 nominal training episodes per task, and the runtime analysis in Appendix A.2 shows that end-to-end inference is dominated by world model computation rather than kNN lookup. For larger-scale real-time deployments, standard extensions such as approximate nearest-neighbor indexing or feature-bank sub-sampling can be applied without changing the underlying anomaly score formulation.

Overall, we believe this work constitutes an important step toward the reliable deployment of reinforcement learning agents in real-world settings. Effective anomaly detection is a prerequisite for safe and robust operation in complex, unstructured environments. Our work demonstrates that world model-based representations offer a strong foundation for meeting this requirement.

Appendix A

A.1 Model Training & Hyper-Parameters

The recurrent state-space model (RSSM) is trained in a self-supervised manner using only nominal episodes, following the architecture of Hafner et al. (2019) with minor modifications to support higher-resolution observations. Training is performed for 200 epochs using full-length episodes, with one episode per batch. We use the Adam optimizer with a learning rate of 2×10^{-4} during world-model training and a KL regularization scaling factor of $\beta = 1.0$. After convergence, the world model parameters are frozen and the model is used exclusively as a feature extractor during downstream evaluation. A complete specification of the model architecture, training configuration, and hyperparameters is provided in Table 5 for reproducibility.

A.2 Inference Time & Used Hardware

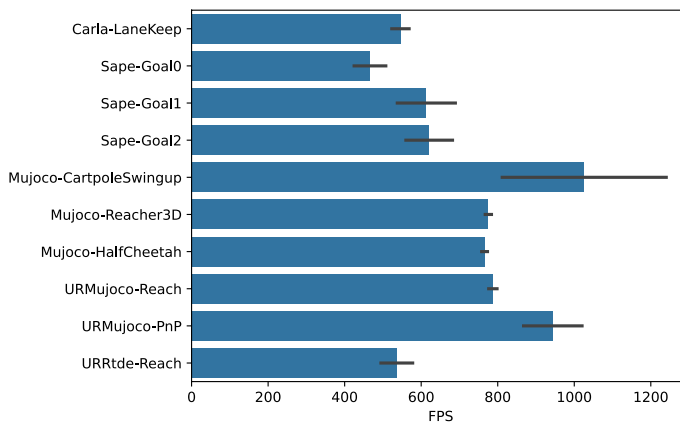
We conducted all our experiments (apart from experiments with PredNet) on a single machine with an NVIDIA GeForce RTX 3090, an AMD Ryzen 9 5950X 16-Core Processor with 62GB of RAM. Inference times of our method on this machine are shown in Fig. 8 and Fig. 9.

A.3 Details on Cross Policy Transfer

To evaluate cross-policy transfer, we trained two additional policies, SAC and TD3, across three different environments until all policies achieved comparable nominal performance (normalized score ≈ 1). Using these trained policies, we constructed new datasets and evaluated the performance of all detectors on each dataset. The results are summarized in Table 6.

Table 5 World Model and Detector Hyperparameters

Category	Hyperparameter (Value)
<i>Rollout and Representation</i>	
Rollout type	Deterministic
Latent feature	Combined (stochastic + deterministic)
<i>RSSM Configuration</i>	
Activation function	ELU
Latent distribution	Normal
Stochastic latent size	50
Deterministic latent size	200
Recurrent hidden size	400
Transition model hidden size	100
Representation model hidden size	100
Observation embedding size	2048
<i>Loss Configuration</i>	
Free nats	1.0
KL loss scale (β)	1.0
<i>Encoder Architecture</i>	
Embedding size	128
Base channel depth	32
Kernel size	4
Stride	2
Activation function	ReLU
Convolutional layers	$C(d) \rightarrow C(2d) \rightarrow C(4d) \rightarrow C(8d)$
<i>Detector Configuration</i>	
Number of neighbors (k)	1
Distance metric	ℓ_1
<i>Optimization</i>	
Optimizer	Adam
Maximum epochs	200
Learning rate	5×10^{-4}
Weight decay	1×10^{-6}

**Fig. 8** Inference Time (fps) of our method on all environments over 3 independent runs

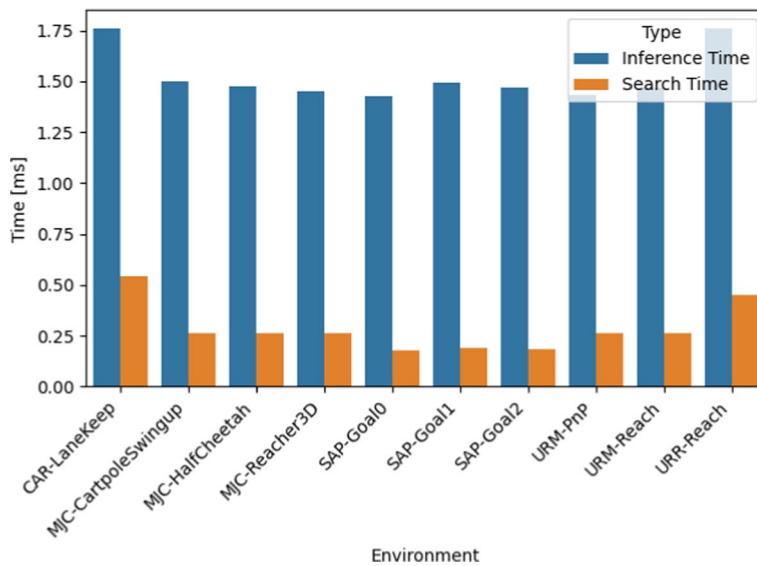


Fig. 9 Inference time, model inference vs. knn search

Table 6 Cross policy transfer
- Normalized scores under anomalous conditions for different rollout policies

Environment	TQC	TD3	SAC
Mujoco-CartpoleSwingup	0.934	0.817	0.816
Sape-Goal0	0.902	0.741	0.887
URMujoco-Reach	0.858	0.738	0.756

Table 6 reports the relative performance of the trained policies under strong anomalous conditions. Although the policies achieve similar success rates in the nominal environment, they exhibit substantially different behaviors when exposed to anomalies. Among the evaluated methods, TQC demonstrates the highest robustness, while TD3 achieves the lowest average rewards in anomalous scenarios.

We then evaluate our approach by training exclusively on data collected from the TQC policy (reference embeddings are also computed using the TQC validation set). The resulting model is then tested on data generated by the SAC and TD3 policies. The corresponding results are reported in Table 2.

A.4 Evaluation Environments

This work builds on the evaluation environments from Anomaly-Gym (Haider et al., 2026), which is work currently under review elsewhere. A copy of the manuscript for Anomaly-Gym is provided in the supplementary material of this submission.

Appendix B Detailed Results Tables

See Table 7.

Table 7 Detailed results for all detectors and environments

Env	Strength	Detector	AUROC		AUPR		FPR95	
			Mean	Std	Mean	Std	Mean	Std
CAR-LaneKeep	Strong	AE-Rec	0.49	0.03	0.51	0.02	0.82	0.03
		CLIP-Sim	0.49	0.06	0.49	0.04	0.88	0.04
		DINO-Patch	0.51	0.07	0.52	0.05	0.86	0.04
		ITRM(ours)	0.92	0.12	0.90	0.13	0.17	0.22
		LRDM	0.69	0.10	0.65	0.08	0.75	0.12
		PredNet	0.52	0.03	0.52	0.03	0.80	0.02
		RES-KNN	0.47	0.06	0.50	0.04	0.81	0.04
		RSSM-KLD	0.71	0.17	0.66	0.14	0.57	0.20
		RSSM-PPM	0.81	0.23	0.79	0.19	0.43	0.35
	Tiny	AE-Rec	0.48	0.01	0.52	0.01	0.84	0.01
		CLIP-Sim	0.41	0.02	0.47	0.01	0.91	0.01
		DINO-Patch	0.42	0.01	0.49	0.01	0.90	0.01
		ITRM(ours)	0.85	0.14	0.81	0.14	0.30	0.21
		LRDM	0.65	0.11	0.61	0.07	0.74	0.13
		PredNet	0.49	0.01	0.53	0.01	0.82	0.01
		RES-KNN	0.39	0.02	0.48	0.01	0.87	0.02
		RSSM-KLD	0.60	0.18	0.59	0.13	0.72	0.16
		RSSM-PPM	0.61	0.23	0.64	0.17	0.69	0.26
MJC-CartpoleSwingup	Strong	AE-Rec	0.74	0.12	0.73	0.12	0.73	0.20
		CLIP-Sim	0.66	0.13	0.64	0.11	0.81	0.15
		DINO-Patch	0.49	0.06	0.52	0.04	0.90	0.09
		ITRM(ours)	0.85	0.20	0.87	0.17	0.30	0.25
		LRDM	0.86	0.15	0.83	0.15	0.42	0.31
		PredNet	0.68	0.09	0.61	0.08	0.66	0.07
		RES-KNN	0.77	0.12	0.73	0.10	0.69	0.19
		RSSM-KLD	0.50	0.13	0.50	0.06	0.88	0.07
		RSSM-PPM	0.62	0.27	0.69	0.18	0.73	0.23
	Tiny	AE-Rec	0.51	0.14	0.54	0.07	0.85	0.14
		CLIP-Sim	0.41	0.16	0.50	0.08	0.90	0.12
		DINO-Patch	0.41	0.12	0.49	0.05	0.90	0.10
		ITRM(ours)	0.74	0.32	0.80	0.23	0.40	0.33
		LRDM	0.56	0.20	0.59	0.13	0.83	0.15
		PredNet	0.64	0.07	0.55	0.06	0.61	0.07
		RES-KNN	0.57	0.17	0.57	0.10	0.81	0.18
		RSSM-KLD	0.34	0.07	0.43	0.03	0.94	0.02
		RSSM-PPM	0.40	0.26	0.53	0.18	0.90	0.15
MJC-HalfCheetah	Strong	AE-Rec	0.44	0.02	0.50	0.03	0.98	0.01
		CLIP-Sim	0.53	0.03	0.55	0.03	0.94	0.01
		DINO-Patch	0.53	0.03	0.57	0.03	0.92	0.01
		ITRM(ours)	0.93	0.04	0.93	0.04	0.28	0.18
		LRDM	0.62	0.01	0.62	0.03	0.83	0.03
		PredNet	0.80	0.06	0.71	0.07	0.45	0.10
		RES-KNN	0.58	0.06	0.62	0.06	0.92	0.02
		RSSM-KLD	0.65	0.06	0.71	0.07	0.86	0.04
		RSSM-PPM	0.55	0.04	0.61	0.05	0.94	0.01
	Tiny	AE-Rec	0.47	0.01	0.52	0.01	0.97	0.01
		CLIP-Sim	0.45	0.01	0.51	0.01	0.96	0.00

Table 7 (continued)

Env	Strength	Detector	AUROC		AUPR		FPR95	
			Mean	Std	Mean	Std	Mean	Std
MJC-Reacher3D	Strong	DINO-Patch	0.49	0.01	0.54	0.01	0.94	0.01
		ITRM(ours)	0.72	0.16	0.74	0.14	0.61	0.16
		LRDM	0.59	0.01	0.56	0.02	0.73	0.02
		PredNet	0.96	0.00	0.90	0.02	0.16	0.01
		RES-KNN	0.42	0.02	0.49	0.01	0.97	0.01
		RSSM-KLD	0.41	0.01	0.49	0.01	0.94	0.01
		RSSM-PPM	0.39	0.01	0.49	0.01	0.97	0.01
		AE-Rec	0.68	0.08	0.72	0.05	0.60	0.06
		CLIP-Sim	0.79	0.07	0.69	0.08	0.57	0.08
		DINO-Patch	0.79	0.07	0.70	0.07	0.53	0.06
	Tiny	ITRM(ours)	0.95	0.03	0.92	0.05	0.13	0.08
		LRDM	0.91	0.05	0.85	0.08	0.27	0.11
		PredNet	0.58	0.11	0.60	0.07	0.68	0.12
		RES-KNN	0.79	0.07	0.69	0.08	0.53	0.07
		RSSM-KLD	0.69	0.15	0.64	0.12	0.58	0.17
		RSSM-PPM	0.68	0.10	0.63	0.10	0.79	0.10
		AE-Rec	0.65	0.04	0.71	0.03	0.59	0.02
		CLIP-Sim	0.69	0.01	0.61	0.01	0.67	0.02
		DINO-Patch	0.69	0.01	0.61	0.01	0.60	0.01
		ITRM(ours)	0.88	0.07	0.81	0.09	0.23	0.08
SAP-Goal0	Strong	LRDM	0.81	0.04	0.73	0.06	0.40	0.02
		PredNet	0.62	0.02	0.61	0.02	0.60	0.01
		RES-KNN	0.69	0.02	0.60	0.02	0.64	0.02
		RSSM-KLD	0.41	0.10	0.46	0.06	0.80	0.06
		RSSM-PPM	0.56	0.06	0.53	0.06	0.79	0.04
		AE-Rec	0.83	0.04	0.82	0.05	0.49	0.07
		CLIP-Sim	0.72	0.03	0.80	0.05	0.66	0.04
		DINO-Patch	0.56	0.04	0.74	0.06	0.78	0.04
		ITRM(ours)	0.87	0.07	0.85	0.07	0.27	0.09
		LRDM	0.67	0.08	0.75	0.08	0.67	0.11
	Tiny	PredNet	0.72	0.06	0.77	0.06	0.63	0.11
		RES-KNN	0.65	0.09	0.76	0.08	0.69	0.07
		RSSM-KLD	0.86	0.03	0.85	0.04	0.31	0.06
		RSSM-PPM	0.65	0.02	0.76	0.06	0.69	0.03
		AE-Rec	0.78	0.01	0.77	0.02	0.55	0.02
		CLIP-Sim	0.72	0.01	0.72	0.02	0.63	0.02
		DINO-Patch	0.51	0.02	0.62	0.02	0.78	0.03
		ITRM(ours)	0.73	0.06	0.71	0.06	0.45	0.06
		LRDM	0.55	0.02	0.64	0.01	0.85	0.02
		PredNet	0.63	0.02	0.68	0.02	0.81	0.02
SAP-Goal1	Strong	RES-KNN	0.50	0.02	0.61	0.02	0.76	0.03
		RSSM-KLD	0.81	0.01	0.79	0.01	0.39	0.02
		RSSM-PPM	0.64	0.02	0.67	0.02	0.67	0.03
		AE-Rec	0.86	0.04	0.86	0.06	0.38	0.06
		CLIP-Sim	0.65	0.02	0.78	0.06	0.75	0.04
		DINO-Patch	0.53	0.03	0.77	0.05	0.77	0.03

Table 7 (continued)

Env	Strength	Detector	AUROC		AUPR		FPR95	
			Mean	Std	Mean	Std	Mean	Std
SAP-Goal2	Tiny	ITRM(ours)	0.97	0.02	0.96	0.02	0.08	0.04
		LRDM	0.82	0.06	0.83	0.07	0.41	0.08
		PredNet	0.80	0.06	0.82	0.07	0.45	0.08
		RES-KNN	0.56	0.02	0.75	0.06	0.74	0.03
		RSSM-KLD	0.91	0.04	0.93	0.03	0.22	0.07
		RSSM-PPM	0.61	0.03	0.77	0.06	0.74	0.03
		AE-Rec	0.80	0.06	0.78	0.03	0.48	0.07
		CLIP-Sim	0.64	0.02	0.70	0.03	0.74	0.02
		DINO-Patch	0.55	0.03	0.68	0.01	0.75	0.02
		ITRM(ours)	0.95	0.01	0.93	0.01	0.13	0.02
		LRDM	0.74	0.07	0.74	0.03	0.52	0.09
		PredNet	0.72	0.06	0.72	0.03	0.56	0.08
		RES-KNN	0.57	0.03	0.67	0.02	0.73	0.03
		RSSM-KLD	0.86	0.03	0.90	0.02	0.27	0.03
		RSSM-PPM	0.59	0.02	0.68	0.02	0.75	0.03
	Strong	AE-Rec	0.90	0.03	0.88	0.04	0.30	0.04
		CLIP-Sim	0.57	0.02	0.72	0.05	0.79	0.03
		DINO-Patch	0.47	0.04	0.67	0.05	0.80	0.03
		ITRM(ours)	0.92	0.03	0.88	0.04	0.15	0.03
		LRDM	0.81	0.05	0.81	0.06	0.40	0.05
		PredNet	0.76	0.06	0.78	0.07	0.50	0.06
		RES-KNN	0.52	0.02	0.68	0.06	0.75	0.03
		RSSM-KLD	0.88	0.03	0.86	0.04	0.22	0.04
		RSSM-PPM	0.50	0.04	0.68	0.06	0.84	0.04
	Tiny	AE-Rec	0.84	0.09	0.81	0.04	0.38	0.10
		CLIP-Sim	0.58	0.03	0.65	0.02	0.77	0.04
		DINO-Patch	0.47	0.05	0.58	0.03	0.79	0.03
		ITRM(ours)	0.89	0.01	0.83	0.02	0.17	0.01
		LRDM	0.74	0.07	0.72	0.04	0.49	0.09
		PredNet	0.69	0.06	0.67	0.03	0.58	0.07
		RES-KNN	0.52	0.04	0.59	0.03	0.73	0.04
		RSSM-KLD	0.85	0.01	0.81	0.02	0.26	0.02
		RSSM-PPM	0.51	0.02	0.58	0.02	0.81	0.01
URM-PnP	Strong	AE-Rec	0.47	0.07	0.53	0.03	0.71	0.09
		CLIP-Sim	0.56	0.07	0.57	0.05	0.81	0.04
		DINO-Patch	0.52	0.03	0.55	0.03	0.83	0.03
		ITRM(ours)	0.76	0.14	0.71	0.13	0.41	0.11
		LRDM	0.50	0.05	0.55	0.03	0.70	0.07
		PredNet	0.29	0.19	0.45	0.11	0.86	0.14
		RES-KNN	0.46	0.09	0.52	0.07	0.84	0.06
		RSSM-KLD	0.62	0.06	0.58	0.04	0.53	0.07
		RSSM-PPM	0.43	0.06	0.49	0.03	0.94	0.02
	Tiny	AE-Rec	0.46	0.02	0.53	0.02	0.68	0.04
		CLIP-Sim	0.46	0.03	0.52	0.02	0.83	0.03
		DINO-Patch	0.49	0.01	0.53	0.01	0.82	0.03
		ITRM(ours)	0.64	0.12	0.60	0.09	0.50	0.07

Table 7 (continued)

Env	Strength	Detector	AUROC		AUPR		FPR95	
			Mean	Std	Mean	Std	Mean	Std
URM-Reach	Strong	LRDM	0.47	0.02	0.54	0.02	0.69	0.04
		PredNet	0.20	0.12	0.41	0.06	0.92	0.07
		RES-KNN	0.36	0.04	0.47	0.03	0.86	0.02
		RSSM-KLD	0.59	0.03	0.56	0.01	0.53	0.04
		RSSM-PPM	0.39	0.03	0.47	0.01	0.95	0.01
		AE-Rec	0.61	0.02	0.62	0.02	0.71	0.03
		CLIP-Sim	0.59	0.06	0.60	0.04	0.83	0.06
		DINO-Patch	0.60	0.06	0.60	0.04	0.79	0.07
		ITRM(ours)	0.93	0.04	0.85	0.07	0.18	0.09
		LRDM	0.61	0.03	0.62	0.02	0.68	0.04
	Tiny	PredNet	0.50	0.09	0.55	0.05	0.78	0.07
		RES-KNN	0.60	0.08	0.61	0.05	0.79	0.09
		RSSM-KLD	0.77	0.06	0.69	0.06	0.47	0.09
		RSSM-PPM	0.53	0.02	0.55	0.02	0.87	0.01
		AE-Rec	0.60	0.01	0.62	0.02	0.72	0.01
		CLIP-Sim	0.53	0.04	0.57	0.02	0.88	0.04
		DINO-Patch	0.54	0.03	0.57	0.02	0.84	0.04
		ITRM(ours)	0.91	0.04	0.81	0.05	0.22	0.08
		LRDM	0.60	0.02	0.61	0.02	0.70	0.02
		PredNet	0.45	0.07	0.53	0.04	0.82	0.06
URR-Reach	Strong	RES-KNN	0.51	0.06	0.56	0.03	0.85	0.06
		RSSM-KLD	0.73	0.05	0.65	0.04	0.52	0.08
		RSSM-PPM	0.52	0.01	0.55	0.02	0.88	0.01
		AE-Rec	0.60	0.05	0.61	0.04	0.64	0.06
		CLIP-Sim	0.58	0.04	0.56	0.02	0.83	0.03
		DINO-Patch	0.55	0.04	0.55	0.03	0.83	0.03
		ITRM(ours)	0.85	0.05	0.73	0.08	0.29	0.07
		LRDM	0.64	0.05	0.63	0.04	0.64	0.08
		PredNet	0.43	0.03	0.48	0.02	0.84	0.02
		RSSM-KLD	0.68	0.07	0.62	0.05	0.62	0.09
	Tiny	RSSM-PPM	0.56	0.04	0.52	0.02	0.84	0.04
		AE-Rec	0.59	0.03	0.60	0.02	0.66	0.03
		CLIP-Sim	0.60	0.02	0.56	0.02	0.83	0.02
		DINO-Patch	0.55	0.03	0.54	0.02	0.83	0.04
		ITRM(ours)	0.81	0.05	0.69	0.06	0.35	0.08
		LRDM	0.59	0.05	0.59	0.04	0.69	0.06
		PredNet	0.44	0.03	0.49	0.01	0.82	0.01
		RSSM-KLD	0.61	0.02	0.58	0.01	0.68	0.02
		RSSM-PPM	0.56	0.03	0.53	0.01	0.85	0.02

The mean and standard deviation are reported for each detector, environment, and anomaly strength

The results are grouped by environment and detector

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Data Availability We evaluate our method on the benchmark introduced in: (Anomaly-Gym, <https://github.com/FraunhoferKS/Anomaly-Gym>)

Declarations

Conflict of interest The authors declare no conflict of interest.

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